Dr. Natalia Zajac

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Webstie: https://zajacn.github.io/

Education

Dr. sc. ETH - ETH Zurich, Switzerland **01.09.2016 – 17.12.2020**
Department of Environmental System Science and Eawag, Swiss Federal Institute of Aquatic Science and Technology

- Title of doctoral thesis: *The genome of the trematode parasite Atriophallophorus winterbourni, Blasco-Costa et al., 2019: A macro-and microevolutionary perspective*

M.Sc. thesis - University of Bern, Switzerland **07.09.2015 – 30.05.2016**
Institute of Ecology and Evolution at, with Prof. Laurent Excoffier, funded through Swiss-European Mobility Program

- Title of Master thesis: *Chromosomal inversions in the human genome: prevalence, geographic distribution and selection pressures.*

M.Sc. in Evolutionary Biology - University of Uppsala, Sweden **01.09.2014 – 08.06.2016**
Center of Evolutionary Biology

B.A. in Biological Sciences - University of Oxford, UK **01.10.2011 - 20.06.2014**

- Title of third year dissertation: Characterization of a hybrid zone of the Common Wall Lizard, *Podarcis muralis*, in Italy
- received a Southern Field Studies Book Prize
  final grade: 2.1

IB Diploma, British International School of Cracow **01.09.2009 - 30.06.2011**

Research experience

Postdoctoral Researcher **01.02.2025 – PRESENT**
Max Planck Institute for Evolutionary Biology
Mentors: Nathalie Feiner, Tobias Uller; Collaborator: Joana Meier
Topic: Disentangling reticulate evolution in the Mediterranean Wall Lizards with a super pangenome

Bioinformatics Scientist **01.10.2021 – 31.01.2025**
Functional Genomics Center Zurich (ETH and University of Zurich)
Mentors: Dr. Hubert Rehrauer, Dr. Weihong Qi

Responsibilities:

- Bioinformatics support and data analysis in the area of Next Generation Sequencing with special focus on long read data analysis
  - advice on design of experiments using long read sequencing
  - computational and statistical analysis of next generation sequencing data
  - education and training of users (PhD students and postdocs) in software usage and data interpretation

  - Topics: Population Genomics, Genome Assemblies and Annotations, WGS, bulk RNA-seq, long read scRNA-seq, metagenomics
- Operation and management of data analysis software for long reads data analysis
  - installation of software
- Processing and quality control of raw data of long read platforms
  - continuous automatic or semi-automatic processing of raw instrument data
  - consistent registration and documentation in LIMS

Postdoctoral Researcher **01.04.2021 – 30.09.2021**
Department of Computational Biology, University of Lausanne
Supervisors: Prof. Christophe Dessimoz, Dr. Natasha Glover
Topic: Comparative genomics of parasitic platyhelminthes with a focus on gene duplication and gene synteny

Doctoral Researcher - ETH Zurich **01.09.2016 – 30.03.2021**
Doctor of Science degree at the Department of Environmental System Science and the Institute of Aquatic Ecology, Eawag
Supervisors: Prof. Jukka Jokela, Dr. Hanna Hartikainen
Topic: Genomics of adaptation of *Atriophallophorus winterbourni* trematode parasite to its host, *Potamopyrgus antipodarum*

Master thesis project – University of Bern **07.09.2015 – 30.05.2016**
Institute of Ecology and Evolution
Supervisors: Prof. Laurent Excoffier and Dr. Isabelle Dupanloup Duperret
Topic: Characterizing the prevalence and geographic distribution of common inversions in human populations using known inversions and structural variant databases

Research assistant – Univeristy of Oxford **01.11.2013 – 10.06.2014**
Department of Zoology
Supervisor: Dr. Tobias Uller, Sozos Michaelides
Topic: Genetics of the common wall lizard, *Podarcis muralis*

Teaching

2026
- course instructor at the **EMBO Pangenomics course**
  Material: https://github.com/zajacn/EMBO_pangenomics_course
- Guest lecturer at the **Evolutionary Biology course** at the **Kadir Has University, Istanbul**

2025
- Guest Lecturer at the course Research Cycle in Genomics at **University of Zurich**

2021 – 2025
- **BIO253 at University of Zurich** – Research Cycle in Genomics – Fall Semester
  Block course for Master and Bachelor students from University of Zurich and ETH (3.5 weeks); teaching partner: Dr. Jonas Grossmann. See:
  https://www.news.uzh.ch/en/articles/news/2023/future-of-teaching.html

- **BIO638 at FGCZ** – Next Generation Sequencing Applied to Metagenomics - Spring and Fall Semester (3 day data analysis course for PhD students and Postdocs). See: https://gitlab.bfabric.org/genomics-courses/bio638_2024_june
  Material: https://github.com/zajacn/metagenomics_course_FGCZ

2019

- **Ecology and Evolution**: Term Paper (701-1460-00 A)

Supervised Students

2016 – 2020

- 6 months: Co-supervision of Master Thesis of Julia Vrtilek together with Prof. Jukka Jokela
- Ecology and Evolution: Term Paper (701-1460-00 A) – supervision of Michael Zehnder
- Lab intership (3 months) - supervision of Michael Zehnder and Rebecca Oester

Prizes & Grants

Micro_innovation UZH Teaching Fund (ULF) June 2024

Best poster award (among 6 other students), Biology19

Best junior talk at PhD symposium at Eawag, 2017

Swiss International Mobility Program Scholarship, 2015 – obtained from University of Bern for Master thesis project

Diploma for participating in Diana Science Conference as a Speaker, Feedback Provider and Audience, 2014

Southern Field Studies Book Prize, 2014 – obtained for the best fieldwork project of the year for third year honours research project

David Kirby Memorial Fund 2013 - fund awarded for Honours Research Project

Demyship Scholarship (Magdalen College Oxford) 2012-2013 – awarded to students obtaining a top grade (first) in end of year exams

BP-NRE award, 2012 – scholarship obtained for the Oxford University International Internship Program, awarded to support students undertaking internships in Natural Resources and the Environment

Selected Conference Talks

Talk at the Society of Molecular Biology conference **06.2026**
Disentangling Reticulate Evolution in the Mediterranean Wall Lizards with a Super Pangenome

Talk at the Women in Evolutionary Biology workshop at the Max Planck Institute for Evolutionary Biology **06.2026**
Invited talk at the Wellcome Sanger Institute at the Pangenomics Journal Club **05.2026**
Talk at a Workshop on Pangenomics at the University of Tennessee Health Science Center (UTHSC) **05.2025**
Talk at ABRF **04.2024**
The impact of PCR duplication on RNA-seq data generated using NovaSeq 6000, NovaSeq X, AVITI and G4 sequencers.
Talk at the Zürich Seminars in Bioinformatics **04.2024**
Comparison of Single-cell Long-read and Short-read Transcriptome Sequencing of Patient-derived Organoid Cells of ccRCC: Quality Evaluation of the MAS-ISO-seq Approach.
Talk at the Swiss Institute of Bioinformatics Days **06.2020**
Gene duplication and gain in Atriophallophorus winterbourni and other major parasitic trematodes contributes to adaptation
Talk at the Biology20 Symposium in Fribourg **02.2020**
Gene duplication and gain in the Trematode *Atriophallophorus winterbourni* contributes to adaptation to parasitism

Courses

2021 - PRESENT

SignUp Leadership Program; selected by Institute Director

Max Planck Courses:
- AI for scientific research

Physalia courses:
- Manual Genome Curation with HiC data

Swiss Institute of Bioinformatics Courses:

- Gene Ontology Enrichment Analysis
- Single Cell Transcriptomics
- Version Control with Git
- Docker and Singularity for reproducible research: getting started with containers
- Introduction to Snakemake for reproducible analyses
- Introduction to High-performance Computing
- Avoiding Pitfalls and Misinterpretation during the Taxonomic Profiling of Metagenomes

International Genome Graph Symposium (IGGSy’22)

ECCB/ISMB:

- Microbiome analysis with Megan6
- Exploring nf-core: a best-practice framework for Nextflow pipelines
- Computational approaches for identifying context-specific transcription factors using single-cell multi-omics datasets

Publications

Main:

Evolutionary Genomics of a trematode parasite:

- Zajac, N., F. Feijen, and J. Jokela. 2026. **A Whole-Genome Investigation of Mitonuclear Discordance in the Trematode Parasite *Atriophallophorus winterbourni*.** Molecular Ecology35, no. 10: e70391. https://doi.org/10.1111/mec.70391.
- Feijen, F., Zajac, N., Vorburger, C., Blasco-Costa, I., & Jokela, J. (2022). **Phylogeography and cryptic species structure of a locally adapted parasite in New Zealand.** *Molecular Ecology*, *31*(15), 4112–4126.
- Zajac, N., Zoller, S., Seppälä, K., Moi, D., Dessimoz, C., Jokela, J., Hartikainen, H., & Glover, N. (2021). **Gene duplication and gain in the trematode Atriophallophorus winterbourni contributes to adaptation to parasitism.** *Genome Biology and Evolution*, *13*(3), evab010.
- Zajac, N. (2020) **The genome of the trematode parasite *Atriophallophorus winterbourni*, Blasco-Costa et al., 2019: A macro-and microevolutionary perspective.** ETH Zurich
- Blasco-Costa, I., Seppälä, K., Feijen, F., Zajac, N., Klappert, K., & Jokela, J. (2020**). A new species of Atriophallophorus Deblock & Rosé, 1964 (Trematoda: Microphallidae) described from in vitro-grown adults and metacercariae from Potamopyrgus antipodarum (Gray, 1843)(Mollusca: Tateidae).** *Journal of Helminthology*, *94*, e108.

Evolutionary Genomics of Reptiles:

- Westelius, T., Pranter, R., Stansfield, C., Zajac, N., & Feiner, N. (2026). **Towards understanding the mechanistic basis of a sex-limited color polymorphism.** bioRxiv, 2026-05.
- Feiner, N., Zajac, N., Wen, G., & Uller, T. (2025). **Population and Evolutionary Genomics of Lizards and Snakes**. EcoEvoarxiv.
- Michaelides, S. N., While, G. M., Zajac, N., Aubret, F., Calsbeek, B., Sacchi, R., Zuffi, M. A. L., & Uller, T. (2016). **Loss of genetic diversity and increased embryonic mortality in non-native lizard populations.** *Molecular Ecology*, *25*(17), 4113–4125.
- Michaelides, S., Cornish, N., Griffiths, R., Groombridge, J., Zajac, N., Walters, G. J., Aubret, F., While, G. M., & Uller, T. (2015). **Phylogeography and conservation genetics of the common wall lizard, *Podarcis muralis*, on islands at its northern range.** *PLoS One*, *10*(2), e0117113.
- Michaelides, S. N., While, G. M., Zajac, N., & Uller, T. (2015). **Widespread primary, but geographically restricted secondary, human introductions of wall lizards, *Podarcis muralis*.** *Molecular Ecology*, *24*(11), 2702–2714.
- While, G. M., Michaelides, S., Heathcote, R. J. P., MacGregor, H. E. A., Zajac, N., Beninde, J., Carazo, P., Perez i de Lanuza, G., Sacchi, R., & Zuffi, M. A. L. (2015). **Sexual selection drives asymmetric introgression in wall lizards.** *Ecology Letters*, *18*(12), 1366–1375.

Method benchmarking:

- Zajac, N., Zhang, Q., Bratus-Neuenschwander, A., Qi, W., Bolck, H. A., Karakulak, T., ... & Rehrauer, H. (2025). **Comparison of single-cell long-read and short-read transcriptome sequencing via cDNA molecule matching: quality evaluation of the MAS-ISO-seq approach.** NAR Genomics and Bioinformatics, 7(3), lqaf089.
- Zajac, N., Vlachos, I. S., Sajibu, S., Opitz, L., Wang, S., Chittur, S. V., ... & Aquino, C. (2025). **The impact of PCR duplication on RNAseq data generated using NovaSeq 6000, NovaSeq X, AVITI, and G4 sequencers**. Genome biology, 26(1), 145.

Co-authorship due to bioinformatic analysis in the following publications:

- Iachizzi, M., Zajac, N., Ruiz, J. L., Güller, T., Rabin, R., Schalbetter, S., ... & Richetto, J. (2026). **Probiotic treatment rescues behavioral deficits and gut microbial abnormalities induced by preconceptional stress in mothers and offspring.** *Brain, Behavior, and Immunity*, 106571.
- Dantas, L. O., Candeiro, G. T., Pereira, A. C., Pereira, A. C., Mita, D., Zajac, N., ... & Pinheiro, E. T. (2026). **Metatranscriptomic Insights into Bacterial Activity, Virulence, and Antimicrobial Resistance in the Root Canal Microbiome of Acute Apical Abscesses.** Journal of Endodontics.
- Schwyter, L., Niggli, S., Zajac, N., Poveda, L., Wolski, W. E., Panse, C., ... & Grossmann, J. (2025). **A combination of phenotypic responses and genetic adaptations enables Staphylococcus aureus to withstand inhibitory molecules secreted by Pseudomonas aeruginosa**. Msystems, 10(8), e00262-25.
- Karakulak, T.*, Zajac, N.*, Bolck, H. A., Bratus-Neuenschwander, A., Zhang, Q., Qi, W., ... & Kahraman, A. (2025). **Heterogeneous and novel transcript expression in single cells of patient-derived clear cell renal cell carcinoma organoids**. Genome Research, 35(4), 698. (* - first authors)
- Bolck, H. A., Migh, E., Kriston, A., Zajac, N. H., Kreutzer, S., Totu, T., ... & Moch, H. (2025). **Deep visual multi-omics profiling reveals mechanisms that underly cancer cell differentiation and aggressiveness in clear cell renal cell carcinoma**. bioRxiv, 2025-01.
- Emmenegger, M., Zografou, C., Dai, Y., Hoyt, L. R., Gudneppanavar, R., Chincisan, A., Rehrauer, H., Noé, F. J., Zajac, N., Meisl, G., Schneider, M. M., Nguyen, H., Höpker, K., Knowles, T. P. J., Sospedra, M., Martin, R., Ring, A. M., Leeds, S., Eisenbarth, S. C., … Aguzzi, A. (2024). **The Cystic Fibrosis Transmembrane Regulator Controls Tolerogenic Responses to Food Allergens in Mice and Humans.** *MedRxiv*, 2024.06.17.24309019.
- John, A., Dreifuss, D., Kang, S., Bratus-Neuenschwander, A., Zajac, N., Topolsky, I., Dondi, A., Aquino, C., Julian, T. R., & Beerenwinkel, N. (2024). **Assessing different next-generation sequencing technologies for wastewater-based epidemiology.** *MedRxiv*, 2024.05.22.24306666.
- Mengis, T., Zajac, N., Bernhard, L., Heggli, I., Herger, N., Devan, J., Marcus, R., Brunner, F., Laux, C., Farshad, M., Distler, O., & Dudli, S. (2024**). Intervertebral disc microbiome in Modic changes: Lack of result replication underscores the need for a consensus in low-biomass microbiome analysis**. *JOR SPINE*, *7*(2), e1330.

References

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